5.3.8.2 *Low-speed external user clock generated from an external source*

In bypass mode, the LSE oscillator is switched off and the input pin is directly connected to the LSE clock detector (LSECSS). The external clock signal has to respect the parameters specified in Table 33, as shown also by the waveforms in Figure 16.

Table 33. Low-speed external user clock characteristics

Specified by design and not tested in production.

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
f_{LSE_ext}	User external clock source frequency	External digital/analog clock	-	32.768	1000	kHz
V_{LSEH}	Digital OSC_IN input high level	External digital clock	0.7 V_{DD}	-	V_{DD}	V
V_{LSEL}	OSC32_IN input pin low level voltage	External digital clock	V_{SS}	-	0.3 V_{DD}	V
$t_{w(LSEH)}/t_{w(LSEL)}$	OSC32_IN high or low time	External digital clock	250	-	-	ns
V_{lsw_H}	Analog low swing OSC_IN high level	External analog low swing clock	0.6		1.225	V
V_{lsw_L}	Analog low swing OSC_IN low level	External analog low swing clock	0.35		0.8	V
V_{lswLSE} (V_{LSEH} - V_{LSEL})	Analog low swing OSC_IN peak-to-peak amplitude	External analog low swing clock	0.2		0.875	V
$DuCy_{LSE}$	Analog low swing OSC_IN duty cycle	External analog Low Swing Clock	45	50	55	%
t_{rLSE}/t_{fLSE}	Analog low swing OSC_IN rise and fall time	Externala analog low swing clock 10 % to 90 %	-	100	200	ns

Figure 16. Low-speed external clock source AC timing diagram